

Name: _____

These questions assess your understanding of the material from Chapters 1, 2 of Openstax Chemistry. Please write your name on the sheets of paper that you use. Carefully read each question. For questions with numerical calculations, you must show each major step necessary to find your final answer. For questions without numerical calculations, you must include clear and correct reasoning to support your answer. Remember to report the units and correct number of significant figures in your answers.

1. Classify the following as an element, a compound, a homogeneous mixture, or a heterogeneous mixture: pure aspirin (acetylsalicylic acid). You must explain your answer with definitions.
ANSWER: _____
2. Provide the element symbol that represents helium.
ANSWER: _____
3. A materials scientist monitors a laboratory furnace. The furnace temperature is -6.9°C . Convert this temperature to kelvin. Report your answer with the correct number of significant figures.
ANSWER: _____
4. Ryan measures the density of a liquid reagent in his lab. He finds that the sample has a mass of 5.6 g and a volume of 55.07 mL. What is the density of the sample? Ensure your answer has the correct number of significant figures.
ANSWER: _____
5. Classify the density of lead as a physical property or a chemical property. Explain your answer.
ANSWER: _____
6. An engineer measures the capacity of an electricity-generating plant. The capacity is 433 MW. What is the capacity in watts? Report your answer in standard notation, as a decimal.
ANSWER: _____
7. Find the difference in the following measurements, ensuring that your answer has the correct number of significant figures.

$$0.1634 \text{ mL} - 0.065283 \text{ mL}$$

ANSWER: _____

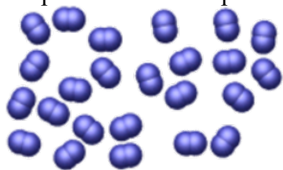
8. As part of her duties as an analytical chemist, Emma analyzes a sample of an unknown pure substance X and finds that it contains 12.2 g of element E_1 and 4.3 g of element E_2 , and no other elements. She then analyzes three other substances that also contain only elements E_1 and E_2 and records the results in a table in her lab notebook, reproduced below. Which of the other three substances is also a pure substance? Explain why you chose your answer.

sample	mass of E_1	mass of E_2
X	12.2 g	4.3 g
A	70.4 g	6.2 g
B	56.4 g	12.3 g
C	52.2 g	4.6 g

ANSWER: _____

9. A pharmacist measures the mass of a powdered drug. The mass is 9066 mg. What is the mass in grams? Report your answer in scientific notation.
ANSWER: _____
10. A laser technician measures the wavelength of light from a new semiconductor laser. The wavelength is $4.1757 \times 10^{-7} \text{ m}$. What is the wavelength in nanometers? Report your answer in standard notation, as a decimal.
ANSWER: _____

11. The following figure is a sketch of a pure substance viewed under a microscope so powerful that it shows individual atoms. Is the pure substance depicted an element, a compound, or a mixture? Explain why you chose your answer.



ANSWER: _____

12. Classify particle count as an intensive or extensive property. You must explain your answer using the definition of the properties.

ANSWER: _____

13. Kevin needs to determine the mass of a sample of ethanol. The density of the sample is $5.1 \frac{\text{g}}{\text{mL}}$ and its volume is 10.37 mL. What is the mass of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____

14. Provide the name and empirical formula of the ionic compound that is formed by the Fe^{3+} cation and the S^{2-} anion.

ANSWER: _____

15. A nurse measures the weight of his patient's medicine. The weight is 0.0083 g. What is the weight in milligrams? Report your answer in standard notation, as a decimal.

ANSWER: _____

16. Identify whether the following describes a random error or a systematic error, and explain your answer: Wind gusts cause repeated outdoor temperature measurements to vary randomly around the true value.

ANSWER: _____

17. A nuclear chemist measures the mass of an artificial element. The mass is 66 pg. What is the mass in grams? Report your answer in scientific notation.

ANSWER: _____

18. Provide the chemical symbol of the most commonly formed ion of potassium and indicate whether it is a cation or an anion. Make sure you include its charge.

ANSWER: _____

19. State the SI base unit used to measure amount of substance and give its symbol.

ANSWER: _____

20. James needs to know the mass of a sample of MgS to 2 significant figures. He puts the sample on a digital scale. This is what the scale shows: 147.05 g. If this measurement is precise enough for James, round it to 2 significant figures and report your answer in scientific notation. Remember to include units in your answer.

ANSWER: _____

21. A medical lab technician measures the mass of cholesterol in a blood sample. The mass is 8.20×10^{-5} g. What is the mass in micrograms? Report your answer in scientific notation.

ANSWER: _____

22. An optician measures the diameter of her patient's pupil. The diameter is 0.0038 m. What is the diameter in millimeters? Report your answer in standard notation, as a decimal.

ANSWER: _____

23. Write the chemical formula of zinc fluoride hexahydrate.

ANSWER: _____

24. Provide the name of the element that the element symbol Ni represents.

ANSWER: _____

25. A geologist measures the height of a mountain. The height is 2.2562 km. What is the height in meters? Report your answer in standard notation, as a decimal.

ANSWER: _____

26. What is the name and symbol of the metric prefix that represents a factor of 10^{-1} ?

ANSWER: _____

27. Add the following measurements, ensuring that your answer has the correct number of significant figures.

$$42.757 \text{ g} + 1.6 \text{ g}$$

ANSWER: _____

28. A cell biologist measures the width of a cell. The width is $68 \mu\text{m}$. What is the width in meters? Report your answer in scientific notation.

ANSWER: _____

29. Classify the following as a homogeneous or heterogeneous mixture: a solution of salt dissolved in water. You must explain your answer and use definitions to support your conclusion.

ANSWER: _____

30. Does the following process produce a new substance? A piece of magnesium ribbon burns in air with a bright white flame, leaving a white powder. You must explain your answer with valid reasoning.

ANSWER: _____

31. Divide the following measurements, ensuring that your answer has the correct number of significant figures.

$$170 \text{ mol} \div 100 \text{ L}$$

ANSWER: _____

32. Classify the following as a physical or chemical change: A banana turning brown after being peeled. Explain your answer.

ANSWER: _____

33. A chemist notes that a reaction is carried out at 167°F . Convert this temperature to degrees Celsius. Round your answer to the nearest integer.

ANSWER: _____

34. A lab technician measures the weight of a reagent. The weight is 4030 g . What is the weight in kilograms? Report your answer in scientific notation.

ANSWER: _____

35. An electrical engineer measures the frequency of a satellite-radio transmission. The frequency is 216 GHz . What is the frequency in hertz? Report your answer in standard notation, as a decimal.

ANSWER: _____

36. Allison is comparing two unknown liquid samples. Sample A has a mass of 15.3 g and a volume of 6.89 mL . Sample B has a mass of $50. \text{ g}$ and a volume of $39. \text{ mL}$. Which sample has the greater density? You must include correct work that supports your conclusion.

ANSWER: _____

37. An arctic ecologist measures the thickness of an ice floe. The thickness is 0.307 m . What is the thickness in centimeters? Report your answer in standard notation, as a decimal.

ANSWER: _____

38. A civil engineer measures the length of an airport runway. The length is 3204 m . What is the length in kilometers? Report your answer in scientific notation.

ANSWER: _____

39. An astrophysicist measures the wavelength of gamma-ray emission from a supernova. The wavelength is $5.39 \times 10^{-14} \text{ m}$. What is the wavelength in femtometers? Report your answer in scientific notation.

ANSWER: _____

40. Julia needs to find the volume of a $17. \text{ g}$ sample of glycerin. The density of the sample is $2.364 \frac{\text{g}}{\text{mL}}$. What is the volume of the sample? Ensure your answer has the correct number of significant figures.

ANSWER: _____